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VLSI MADE EASY

A unique blog which will give you the great interaction to the most advanced enhancements to the new technical trends in Very Large Scale In
The main aim of this blog is to bring you the all available materials, books related to VLSI which would help mostly VLSI Students

VLSI Interview Questions

VLSI FAQs

1. What is metastability?

When setup or hold window is violated in an flip flop then signal attains a unpredictable value or state known as metastability.

2. What is MTBF? What it signifies?

- MTBF-Mean Time Before Failure
- Average time to next failure

3. How chance of metastable state failure can be reduced?

- Lowering clock frequency
- Lowering data speed
- Using faster flip flop

4. What are the advantages of using synchronous reset ?

- No metastability problem with synchronous reset (provided recovery and removal time for reset is taken care).
- Simulation of synchronous reset is easy.

5. What are the disadvantages of using synchronous reset ?

- Synchronous reset is slow.
- Implementation of synchronous reset requires more number of gates compared to asynchronous reset design.
- An active clock is essential for a synchronous reset design. Hence you can expect more power consumption.

6. What are the advantages of using asynchronous reset ?

- Implementation of asynchronous reset requires less number of gates compared to synchronous reset design.
- Asynchronous reset is fast.
- Clocking scheme is not necessary for an asynchronous design. Hence design consumes less power. Asynchronous design style is also one of the latest design options to achieve low power. Design community is scrathing their head over asynchronous design possibilities.

7. What are the disadvantages of using asynchronous reset ?

- Metastability problems are main concerns of asynchronous reset scheme (design).
- Static timing analysis and DFT becomes difficult due to asynchronous reset.

8. What are the 3 fundamental operating conditions that determine the delay characteristics of gate? How operating conditions affect gate delay?

- Process
- Voltage
- Temperature

9. Is verilog/VHDL is a concurrent or sequential language?

- Verilog and VHDL both are concurrent languages.
- Any hardware descriptive language is concurrent in nature.

10. In a system with insufficient hold time, will slowing down the clock frequency help?

- No.
- Making data path slower can help hold time but it may result in setup violation.

11. In a system with insufficient setup time, will slowing down the clock frequency help?

- Yes.
- Making data path faster can also help setup time but it may result in hold violation.

Physical Design Objective Type of Questions and Answers

- **1) Chip utilization depends on ___.**

a. Only on standard cells b. Standard cells and macros c. Only on macros d. Standard cells macros and IO pads

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- ASIC Design (2
- BOOKS (1)
- Digital Design (
- Low Power Des
- Physical Desig
- STA (1)
- STATIC TIMING

Blog Archive

- ► 2011 (13)
- ► 2009 (1)

- **2) In Soft blockages _____ cells are placed.**

a. Only sequential cells b. No cells c. Only Buffers and Inverters d. Any cells

- **3) Why we have to remove scan chains before placement?**

a. Because scan chains are group of flip flop b. It does not have timing critical path c. It is series of flip flop connected in FIFO d. None

- **4) Delay between shortest path and longest path in the clock is called _____.**

a. Useful skew b. Local skew c. Global skew d. Slack

- **5) Cross talk can be avoided by ____.**

a. Decreasing the spacing between the metal layers b. Shielding the nets c. Using lower metal layers d. Using long nets

- **6) Prerouting means routing of _____.**

a. Clock nets b. Signal nets c. IO nets d. PG nets

- **7) Which of the following metal layer has Maximum resistance?**

a. Metal1 b. Metal2 c. Metal3 d. Metal4

- **8) What is the goal of CTS?**

a. Minimum IR Drop b. Minimum EM c. Minimum Skew d. Minimum Slack

- **9) Usually Hold is fixed ____.**

a. Before Placement b. After Placement c. Before CTS d. After CTS

- **10) To achieve better timing _____ cells are placed in the critical path.**

a. HVT b. LVT c. RVT d. SVT

- **11) Leakage power is inversely proportional to ____.**

a. Frequency b. Load Capacitance c. Supply voltage d. Threshold Voltage

- **12) Filler cells are added ____.**

a. Before Placement of std cells b. After Placement of Std Cells c. Before Floor planning d. Before Detail Routing

- **13) Search and Repair is used for ____.**

a. Reducing IR Drop b. Reducing DRC c. Reducing EM violations d. None

- **14) Maximum current density of a metal is available in ____.**

a. .lib b. .v c. .tf d. .sdc

- **15) More IR drop is due to ____.**

a. Increase in metal width b. Increase in metal length c. Decrease in metal length d. Lot of metal layers

- **16) The minimum height and width a cell can occupy in the design is called as ____.**

a. Unit Tile cell b. Multi heighten cell c. LVT cell d. HVT cell

- **17) CRPR stands for ____.**

a. Cell Convergence Pessimism Removal b. Cell Convergence Preset Removal c. Clock Convergence Pessimism Removal d. Clock Convergence Preset Removal

- **18) In OCV timing check, for setup time, ____.**

a. Max delay is used for launch path and Min delay for capture path b. Min delay is used for launch path and Max delay for capture path c. Both Max delay is used for launch and Capture path d. Both Min delay is used for both Capture and Launch paths

- **19) "Total metal area and(or) perimeter of conducting layer / gate to gate area" is called ____.**

a. Utilization b. Aspect Ratio c. OCV d. Antenna Ratio

- **20) The Solution for Antenna effect is ____.**

a. Diode insertion b. Shielding c. Buffer insertion d. Double spacing

- **21) To avoid cross talk, the shielded net is usually connected to ____.**

a. VDD b. VSS c. Both VDD and VSS d. Clock

- **22) If the data is faster than the clock in Reg to Reg path ____ violation may come.**

a. Setup b. Hold c. Both d. None

- **23) Hold violations are preferred to fix ____.**

a. Before placement b. After placement c. Before CTS d. After CTS

- **24) Which of the following is not present in SDC ____?**

a. Max tran b. Max cap c. Max fanout d. Max current density

- **25) Timing sanity check means (with respect to PD)____.**

a. Checking timing of routed design with out net delays b. Checking Timing of placed design with net delays c. Checking Timing of unplaced design without net delays d. Checking Timing of routed design with net delays

- **26) Which of the following is having highest priority at final stage (post routed) of the design ____?**

a. Setup violation b. Hold violation c. Skew d. None

- **27) Which of the following is best suited for CTS?**

a. CLKBUF b. BUF c. INV d. CLKINV

- **28) Max voltage drop will be there at(with out macros) ____.**

a. Left and Right sides b. Bottom and Top sides c. Middle d. None

- **29) Which of the following is preferred while placing macros ____?**

a. Macros placed center of the die b. Macros placed left and right side of die c. Macros placed bottom and top sides of die d. Macros placed based on connectivity of the I/O

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- **30) Routing congestion can be avoided by ____.**

a. placing cells closer b. Placing cells at corners c. Distributing cells d. None

- **31) Pitch of the wire is ____.**

a. Min width b. Min spacing c. Min width - min spacing d. Min width + min spacing

- **32) In Physical Design following step is not there ____.**

a. Floorplaning b. Placement c. Design Synthesis d. CTS

- **33) In technology file if 7 metals are there then which metals you will use for power?**

a. Metal1 and metal2 b. Metal3 and metal4 c. Metal5 and metal6 d. Metal6 and metal7

- **34) If metal6 and metal7 are used for the power in 7 metal layer process design then which metals you will use for clock ?**

a. Metal1 and metal2 b. Metal3 and metal4 c. Metal4 and metal5 d. Metal6 and metal7

- **35) In a reg to reg timing path Tclocktoq delay is 0.5ns and TCombodelay is 5ns and Tsetup is 0.5ns then the clock period should be ____.**

a. 1ns b. 3ns c. 5ns d. 6ns

- **36) Difference between Clock buff/inverters and normal buff/inverters is __.**

a. Clock buff/inverters are faster than normal buff/inverters b. Clock buff/inverters are slower than normal buff/inverters c. Clock buff/inverters are having equal rise and fall times with high drive strengths compare to normal buff/inverters d. Normal buff/inverters are having equal rise and fall times with high drive strengths compare to Clock buff/inverters.

- **37) Which configuration is more preferred during floorplaning ?**

a. Double back with flipped rows b. Double back with non flipped rows c. With channel spacing between rows and no double back d. With channel spacing between rows and double back

- **38) What is the effect of high drive strength buffer when added in long net?**

a. Delay on the net increases b. Capacitance on the net increases c. Delay on the net decreases d. Resistance on the net increases.

- **39) Delay of a cell depends on which factors ?**

a. Output transition and input load b. Input transition and Output load c. Input transition and Output transition d. Input load and Output Load.

- **40) After the final routing the violations in the design ____.**

a. There can be no setup, no hold violations b. There can be only setup violation but no hold c. There can be only hold violation not Setup violation d. There can be both violations.

- **41) Utilisation of the chip after placement optimisation will be ____.**

a. Constant b. Decrease c. Increase d. None of the above

- **42) What is routing congestion in the design?**

a. Ratio of required routing tracks to available routing tracks b. Ratio of available routing tracks to required routing tracks c. Depends on the routing layers available d. None of the above

- **43) What are preroutes in your design?**

a. Power routing b. Signal routing c. Power and Signal routing d. None of the above.

- **44) Clock tree doesn't contain following cell ____.**

a. Clock buffer b. Clock Inverter c. AOI cell d. None of the above

- **Answers:**

1)b 2)c 3)b 4)c 5)b 6)d 7)a 8)c 9)d 10)b 11)d 12)d 13)b 14)c 15)b 16)a 17)c 18)a 19)d 20)a 21)b 22)b 23)d 24)d 25)c 26)b 27)a 28)c 29)d 30)c 31)d 32)c 33)d 34)c 35)d 36)c 37)a 38)c 39)b 40)d 41)c 42)a 43)a 44)c

CMOS Design Interview Questions

Below are the important VLSI CMOS interview questions. This set of interview questions may be updated in future. Answers will be posted one by one as and when i prepare them ! Readers are encouraged to post answers in comment section. Here we go.....

· Draw Vds-Ids curve for an MOSFET. How it varies with a) increasing Vgs b)velocity saturation c)Channel length modulation d)W/L ratio.
· What is body effect? Write mathematical expression? Is it due to parallel or serial connection of MOSFETs?
· What is latch-up in CMOS design and what are the ways to prevent it?
· What is Noise Margin? Explain with the help of Inverter.
· What happens to delay if you increase load capacitance?
· Give the various techniques you know to minimize power consumption for CMOS logic?
· What happens when the PMOS and NMOS are interchanged with one another in an inverter?
· What is body effect?
· Why is NAND gate preferred over NOR gate for fabrication?
· What is Noise Margin? Explain the procedure to determine Noise Margin
· Explain sizing of the inverter?
· How do you size NMOS and PMOS transistors to increase the threshold voltage?
· What happens to delay if we include a resistance at the output of a CMOS circuit?
· What are the limitations in increasing the power supply to reduce delay?
· How does Resistance of the metal lines vary with increasing thickness and increasing length?
· What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus?
· Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?
· Give the expression for CMOS switching power dissipation?
· Why is the substrate in NMOS connected to ground and in PMOS to VDD?
· What is the fundamental difference between a MOSFET and BJT ?
· Which transistor has higher gain- BJT or MOS and why?
· Why PMOS and NMOS are sized equally in a Transmission Gates?
· What is metastability? When/why it will occur? What are the different ways to avoid this?

· Explain zener breakdown and avalanche breakdown?
* What happens if Vds is increased over saturation?
· In the I-V characteristics curve, why is the saturation curve flat or constant?
· What happens if a resistor is added in series with the drain in a CMOS transistor?
· What are the different regions of operation in a CMOS transistor?
· What are the effects of the output characteristics for a change in the beta (β) value?
· What is the effect of body bias?
· What is hot electron effect and how can it be eliminated?
· What is channel length modulation?
· What is the effect of temperature on threshold voltage?
· What is the effect of temperature on mobility?
· What is the effect of gate voltage on mobility?
· What are the different types of scaling?
· What is stage ratio?
· What is charge sharing on a bus?
· What is electron migration and how can it be eliminated?
· Can both PMOS and NMOS transistors pass good 1 and good 0? Explain.
· Why is only NMOS used in pass transistor logic?
· What are the different methodologies used to reduce the charge sharing in dynamic logic?
· What are setup and hold time violations? How can they be eliminated?
· Explain the operation of basic SRAM and DRAM.
· Which ones take more time in SRAM: Read operation or Write operation? Why?
· What is meant by clock race?
· What is meant by single phase and double phase clocking?
· If given a choice between NAND and NOR gates, which one would you pick? Explain.
· Explain the origin of the various capacitances in the CMOS transistor and the physical reasoning behind it.
· Why should the number of CMOS transistors that are connected in series be reduced?
· What is charge sharing between bus and memory element?
· What is crosstalk and how can it be avoided?
· Realize an XOR gate using NAND gate.
· What are the advantages and disadvantages of Bi-CMOS process?
· Draw an XOR gate with using minimum number of transistors and explain the operation.
· What are the critical parameters in a latch and flip-flop?
· What is the significance of sense amplifier in an SRAM?
· Explain Domino logic.
· What are the advantages of depletion mode devices over the enhancement mode devices?
· How can the rise and fall times in an inverter be equated?
· What is meant by leakage current?
· Realize an OR gate using NAND gate.
· Realize an NAND gate using a 2:1 multiplexer.
· Realize an NOR gate using a 2:1 multiplexer.
· Draw the layout of a simple inverter.
· What are the substrates of PMOS and NMOS transistors connected to and explain the results if the connections are interchanged with the other.
· What are repeaters?
· What is tunneling problem?
· What is meant by negative biased instability and how can it be avoided?
· What is Elmore delay algorithm?
· What is meant by metastability?
· What is the effect of Vdd on delay?
· What is the effect of delay, rise and fall times with increase in load capacitance?
· What is the value of mobility of electrons?
· What is value of mobility of holes?
· Give insights of an inverter. Draw Layout. Explain the working.
* Give insights of a 2 input NOR gate. Draw Layout. Explain the working.
· Give insights of a 2 input NAND gate. Draw layout. Explain the working?
· Implement F= not (AB+CD) using CMOS gates.
· What is a pass gate. Explain the working?
· Why do we need both PMOS and NMOS transistors to implement a pass gate?
· What does the above code synthesize to?
· Draw cross section of a PMOS transistor.
· Draw cross section of an NMOS transistor.
· What is a D-latch?
· Implement D flip-flop with a couple of latches?
· Implement a 2 input AND gate using transmission gate?
· Explain various adders and difference between them?
· How can you construct both PMOS and NMOS on a single substrate?
· What happens when the gate oxide is very thin?
· What is SPICE?
· What are the differences between IRSIM and SPICE?
· What are the differences between netlist of HSPICE and Spectre?
· Implement F = AB+C using CMOS gates?
· What is hot electron effect?
· Define threshold voltage?
· List out the factors affecting power consumption on a chip?
· What r the phenomenon which come into play when the devices are scaled to the sub-micron lengths?
· What is clock feed through?
· Implement an Inverter using a single transistor?
· What is Fowler-Nordheim Tunneling?
· Which gate is normally preferred while implementing circuits using CMOS logic, NAND or NOR? Why?

· Draw the Differential Sense Amplifier and explain its working. How to size this circuit?
· What happens if we use an Inverter instead of the Differential Sense Amplifier?
· Draw the SRAM Write Circuitry
· How did you arrive at sizes of transistor in SRAM?
· How does the size of PMOS pull up transistors for bit and bitbar lines affect SRAM's performance?
· What is the critical path in a SRAM?
· Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?
· Give a big picture of the entire SRAM layout showing placements of SRAM cells, row decoders, column decoders, read circuit, write circuit and buffers.
· In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?

Design For Test-DFT

In scan chains if some flip flops are +ve edge triggered and remaining flip flops are -ve edge triggered how it behaves?

Answer:
For designs with both positive and negative clocked flops, the scan insertion tool will always route the scan chain so that the negative clocked flops come before the positive edge flops in the chain. This avoids the need of lockup latch.
For the same clock domain the negedge flops will always capture the data just captured into the posedge flops on the posedge of the clock.
For the multiple clock domains, it all depends upon how the clock trees are balanced. If the clock domains are completely asynchronous, ATPG has to mask the receiving flops.

What you mean by scan chain reordering?

Answer1:
Based on timing and congestion the tool optimally places standard cells. While doing so, if scan chains are detached, it can break the chain ordering (which is done by a scan insertion tool like DFT compiler from Synopsis and can reorder to optimize it.... it maintains the number of flops in a chain.

Answer2:
During placement, the optimization may make the scan chain difficult to route due to congestion. Hence the tool will re-order the chain to reduce congestion.
This sometimes increases hold time problems in the chain. To overcome these buffers may have to be inserted into the scan path. It may not be able to maintain the scan chain length exactly. It cannot swap cell from different clock domains.
Because of scan chain reordering patterns generated earlier is of no use. But this is not a problem as ATPG can be redone by reading the new net list.

what are the differences between SIMULATION and SYNTHESIS

Simulation <= verify your design.

synthesis <= Check for your timing

Simulation is used to verify the functionality of the circuit.. a)Functional Simulation:study of ckt's operation independent of timing parameters and gate delays. b) Timing Simulation :study including estimated delays, verify setup,hold and other timing requirements of devices like flip flops are met.

Synthesis:One of the foremost in back end steps where by synthesizing is nothing but converting VHDL or VERILOG description to a set of primitives(equations as in CPLD) or components(as in FPGA'S)to fit into the target technology.Basically the synthesis tools convert the design description into equations or components

Can u tell me the differences between latches & flipflops?

There are 2 types of circuits:

1. Combinational

2. Sequential

Latches and flipflops both come under the category of "sequential circuits", whose output depends not only on the current inputs, but also on previous inputs and outputs. Difference: Latches are level-sensitive, whereas, FF are edge sensitive. By edge sensitive, I mean O/p changes only when there is a clock transition.(from 1 to 0, or from 0 to 1)

Example: In a flipflop, inputs have arrived on the input lines at time= 2 seconds. But, output won't change immediately. At time = 3 seconds, clock transition takes place. After that, O/P will change.

Flip-flops are of 2 types:

1.Positive edge triggered

2. negative edge triggered

1)fllipflops take twice the nymber of gates as latches

2) so automatically delay is more for flipflops

3)power consumption is also more

latch does not have a clock signal, whereas a flip-flop always does.

What is slack?

The slack is the time delay difference from the expected delay(1/clock) to the actual delay in a particular path.

Slack may be +ve or -ve.

Equivalence between VHDL and C?

There is concept of understanding in C there is structure.Based upon requirement structure provide facility to store collection of different data types.

In VHDL we have direct access to memory so instead of using pointer in C (and member of structure) we can write interface store data in memory and access it.

RTL and Behavioral

Register transfer language means there should be data flow between two registers and logic is in between them for end registers

data should flow.

Behavioral means how hardware behave determine the exact way it works we write using HDL syntax.For complex projects it is better mixed approach or more behavioral is used.

VHDL QUESTIONS

1. What is the difference between using direct instantiations and component ones except that you need to declare the component?
2. What is the use of BLOCKS?
3. What is the use of PROCEDURES?
4. What is the usage of using more then one architecture in an entity?
5. What is a D-latch? Write the VHDL Code for it?
6. Implement D flip-flop with a couple of latches? Write a VHDL Code for a D flip-flop?
7. Differences between Signals and Variables in VHDL? If the same code is written using Signals and Variables what does it synthesize to?
8. Differences between functions and Procedures in VHDL?
9. Explain the concept of a Clock Divider Circuit? Write a VHDL code for the same?

Digital Design interview questions:

1. Give two ways of converting a two input NAND gate to an inverter
2. Given a circuit, draw its exact timing response. (I was given a Pseudo Random Signal Generator; you can expect any sequential ckt)
3. What are set up time & hold time constraints? What do they signify? Which one is critical for estimating maximum clock frequency of a circuit?
4. Give a circuit to divide frequency of clock cycle by two
5. Design a divide-by-3 sequential circuit with 50% duty circle. (Hint: Double the Clock)
6. Suppose you have a combinational circuit between two registers driven by a clock. What will you do if the delay of the combinational circuit is greater than your clock signal? (You can't resize the combinational circuit transistors)
7. The answer to the above question is breaking the combinational circuit and pipelining it. What will be affected if you do this?
8. What are the different Adder circuits you studied?
9. Give the truth table for a Half Adder. Give a gate level implementation of the same.
10. Draw a Transmission Gate-based D-Latch.
11. Design a Transmission Gate based XOR. Now, how do you convert it to XNOR? (Without inverting the output)
12. How do you detect if two 8-bit signals are same?
13. How do you detect a sequence of "1101" arriving serially from a signal line?
14. Design any FSM in VHDL or Verilog.

Intel interview questions

The following questions are used for screening the candidates during the first interview. The questions apply mostly to fresh college grads pursuing an engineering career at Intel.

1. Have you studied buses? What types?
2. Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an instruction in a 5 stage machine? What is the throughput of this machine ?
3. How many bit combinations are there in a byte?
4. For a single computer processor computer system, what is the purpose of a processor cache and describe its operation?
5. Explain the operation considering a two processor computer system with a cache for each processor.
6. What are the main issues associated with multiprocessor caches and how might you solve them?
7. Explain the difference between write through and write back cache.
8. Are you familiar with the term MESI?
9. Are you familiar with the term snooping?
10. Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.
11. In what cases do you need to double clock a signal before presenting it to a synchronous state machine?
12. You have a driver that drives a long signal & connects to an input device. At the input device there is either overshoot, undershoot or signal threshold violations, what can be done to correct this problem?
13. What are the total number of lines written by you in C/C++? What is the most complicated/valuable program written in C/C++?
14. What compiler was used?
15. What is the difference between = and == in C?
16. Are you familiar with VHDL and/or Verilog?
17. What types of CMOS memories have you designed? What were their size? Speed?
18. What work have you done on full chip Clock and Power distribution? What process technology and budgets were used?
19. What types of I/O have you designed? What were their size? Speed? Configuration? Voltage requirements?
20. Process technology? What package was used and how did you model the package/system? What parasitic effects were considered?
21. What types of high speed CMOS circuits have you designed?

22. What transistor level design tools are you proficient with? What types of designs were they used on?
23. What products have you designed which have entered high volume production?
24. What was your role in the silicon evaluation/product ramp? What tools did you use?
25. If not into production, how far did you follow the design and why did not you see it into production?

VLSI Design Interview questions

1. Explain why & how a MOSFET works
2. Draw Vds-Ids curve for a MOSFET. Now, show how this curve changes (a) with increasing Vgs (b) with increasing transistor width (c) considering Channel Length Modulation
3. Explain the various MOSFET Capacitances & their significance
4. Draw a CMOS Inverter. Explain its transfer characteristics
5. Explain sizing of the inverter
6. How do you size NMOS and PMOS transistors to increase the threshold voltage?
7. What is Noise Margin? Explain the procedure to determine Noise Margin
8. Give the expression for CMOS switching power dissipation
9. What is Body Effect?
10. Describe the various effects of scaling
11. Give the expression for calculating Delay in CMOS circuit
12. What happens to delay if you increase load capacitance?
13. What happens to delay if we include a resistance at the output of a CMOS circuit?
14. What are the limitations in increasing the power supply to reduce delay?
15. How does Resistance of the metal lines vary with increasing thickness and increasing length?
16. You have three adjacent parallel metal lines. Two out of phase signals pass through the outer two metal lines. Draw the waveforms in the center metal line due to interference. Now, draw the signals if the signals in outer metal lines are in phase with each other
17. What happens if we increase the number of contacts or via from one metal layer to the next?
18. Draw a transistor level two input NAND gate. Explain its sizing (a) considering Vth (b) for equal rise and fall times
19. Let A & B be two inputs of the NAND gate. Say signal A arrives at the NAND gate later than signal B. To optimize delay, of the two series NMOS inputs A & B, which one would you place near the output?
20. Draw the stick diagram of a NOR gate. Optimize it
21. For CMOS logic, give the various techniques you know to minimize power consumption
22. What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus
23. Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?
24. In the design of a large inverter, why do we prefer to connect small transistors in parallel (thus increasing effective width) rather than lay out one transistor with large width?
25. Given a layout, draw its transistor level circuit. (I was given a 3 input AND gate and a 2 input Multiplexer. You can expect any simple 2 or 3 input gates)
26. Give the logic expression for an AOI gate. Draw its transistor level equivalent. Draw its stick diagram
27. Why don't we use just one NMOS or PMOS transistor as a transmission gate?
28. For a NMOS transistor acting as a pass transistor, say the gate is connected to VDD, give the output for a square pulse input going from 0 to VDD
29. Draw a 6-T SRAM Cell and explain the Read and Write operations
30. Draw the Differential Sense Amplifier and explain its working. Any idea how to size this circuit? (Consider Channel Length Modulation)
31. What happens if we use an Inverter instead of the Differential Sense Amplifier?
32. Draw the SRAM Write Circuitry
33. Approximately, what were the sizes of your transistors in the SRAM cell? How did you arrive at those sizes?
34. How does the size of PMOS Pull Up transistors (for bit & bit- lines) affect SRAM's performance?
35. What's the critical path in a SRAM?
36. Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?
37. Give a big picture of the entire SRAM Layout showing your placements of SRAM Cells, Row Decoders, Column Decoders, Read Circuit, Write Circuit and Buffers
38. In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?
39. How can you model a SRAM at RTL Level?
40. Whatï¿½s the difference between Testing & Verification?
41. For an AND-OR implementation of a two input Mux, how do you test for Stuck-At-0 and Stuck-At-1 faults at the internal nodes? (You can expect a circuit with some redundant logic)
42. What is Latch Up? Explain Latch Up with cross section of a CMOS Inverter. How do you avoid Latch Up?

Verilog Interview Questions

- What is the difference between $display and $monitor and $write and $strobe?
- What is the difference between code-compiled simulator and normal simulator?

- What is the difference between wire and reg?
- What is the difference between blocking and non-blocking assignments?
- What is the significance Timescale directivbe?
- What is the difference between bit wise, unary and logical operators?
- What is the difference between task and function?
- What is the difference between casex, casez and case statements?
- Which one preferred-casex or casez?
- For what is defparam used?
- What is the difference between "= =" and "= = =" ?
- What is a compiler directive like 'include' and 'ifdef'?
- Write a verilog code to swap contents of two registers with and without a temporary register?
- What is the difference between inter statement and intra statement delay?
- What is delta simulation time?
- What is difference between Verilog full case and parallel case?
- What you mean by inferring latches?
- How to avoid latches in your design?
- Why latches are not preferred in synthesized design?
- How blocking and non blocking statements get executed?
- Which will be updated first: is it variable or signal?
- What is sensitivity list?
- If you miss sensitivity list what happens?
- In a pure combinational circuit is it necessary to mention all the inputs in sensitivity disk? If yes, why? If not, why?
- In a pure sequential circuit is it necessary to mention all the inputs in sensitivity disk? If yes, why? If not, why?
- What is general structure of Verilog code you follow?
- What are the difference between Verilog and VHDL?
- What are system tasks?
- List some of system tasks and what are their purposes?
- What are the enhancements in Verilog 2001?
- Write a Verilog code for synchronous and asynchronous reset?
- What is pli? why is it used?
- What is file I/O?
- What is difference between freeze deposit and force?
- Will case always infer priority register? If yes how? Give an example.
- What are inertial and transport delays ?
- What does `timescale 1 ns/ 1 ps' signify in a verilog code?
- How to generate sine wav using verilog coding style?
- How do you implement the bi-directional ports in Verilog HDL?
- How to write FSM is verilog?
- What is verilog case (1)?
- What are Different types of Verilog simulators available?
- What is Constrained-Random Verification ?
- How can you model a SRAM at RTL Level?

Physical Design Questions and Answers

- I am getting several emails requesting answers to the questions posted in this blog. But it is very difficult to provide detailed answer to all questions in my available spare time. Hence i decided to give "short and sweet" one line answers to the questions so that readers can immediately benefited. Detailed answers will be posted in later stage.I have given answers to some of the physical design questions here. Enjoy !

What parameters (or aspects) differentiate Chip Design and Block level design?

- Chip design has I/O pads; block design has pins.
- Chip design uses all metal layes available; block design may not use all metal layers.
- Chip is generally rectangular in shape; blocks can be rectangular, rectilinear.
- Chip design requires several packaging; block design ends in a macro.

How do you place macros in a full chip design?

- First check flylines i.e. check net connections from macro to macro and macro to standard cells.
- If there is more connection from macro to macro place those macros nearer to each other preferably nearer to core boundaries.
- If input pin is connected to macro better to place nearer to that pin or pad.
- If macro has more connection to standard cells spread the macros inside core.
- Avoid criscross placement of macros.
- Use soft or hard blockages to guide placement engine.

Differentiate between a Hierarchical Design and flat design?

- Hierarchial design has blocks, subblocks in an hierarchy; Flattened design has no subblocks and it has only leaf cells.
- Hierarchical design takes more run time; Flattened design takes less run time.

Which is more complicated when u have a 48 MHz and 500 MHz clock design?

- 500 MHz; because it is more constrained (i.e.lesser clock period) than 48 MHz design.

Name few tools which you used for physical verification?

- Herculis from Synopsys, Caliber from Mentor Graphics.

What are the input files will you give for primetime correlation?

- Netlist, Technology library, Constraints, SPEF or SDF file.

If the routing congestion exists between two macros, then what will you do?

- Provide soft or hard blockage

How will you decide the die size?

- By checking the total area of the design you can decide die size.

If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?

- Poly

If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?

- Because top two metal layers are required for global routing in chip design. If top metal layers are also used in block level it will create routing blockage.

In your project what is die size, number of metal layers, technology, foundry, number of clocks?

- Die size: tell in mm eg. 1mm x 1mm ; remeber 1mm=1000micron which is a big size !!
- Metal layers: See your tech file. generally for 90nm it is 7 to 9.
- Technology: Again look into tech files.
- Foundry:Again look into tech files; eg. TSMC, IBM, ARTISAN etc
- Clocks: Look into your design and SDC file !

How many macros in your design?

- You know it well as you have designed it ! A SoC (System On Chip) design may have 100 macros also !!!!

What is each macro size and number of standard cell count?

- Depends on your design.

What are the input needs for your design?

- For synthesis: RTL, Technology library, Standard cell library, Constraints
- For Physical design: Netlist, Technology library, Constraints, Standard cell library

What is SDC constraint file contains?

- Clock definitions
- Timing exception-multicycle path, false path
- Input and Output delays

How did you do power planning?
How to calculate core ring width, macro ring width and strap or trunk width?
How to find number of power pad and IO power pads?
How the width of metal and number of straps calculated for power and ground?

- Get the total core power consumption; get the metal layer current density value from the tech file; Divide total power by number sides of the chip; Divide the obtained value from the current density to get core power ring width. Then calculate number of straps using some more equations. Will be explained in detail later.

How to find total chip power?

- Total chip power=standard cell power consumption,Macro power consumption pad power consumption.

What are the problems faced related to timing?

- Prelayout: Setup, Max transition, max capacitance
- Post layout: Hold

How did you resolve the setup and hold problem?

- Setup: upsize the cells
- Hold: insert buffers

In which layer do you prefer for clock routing and why?

- Next lower layer to the top two metal layers(global routing layers). Because it has less resistance hence less RC delay.

If in your design has reset pin, then it'll affect input pin or output pin or both?

- Output pin.

During power analysis, if you are facing IR drop problem, then how did you avoid?

- Increase power metal layer width.
- Go for higher metal layer.
- Spread macros or standard cells.
- Provide more straps.

Define antenna problem and how did you resolve these problem?

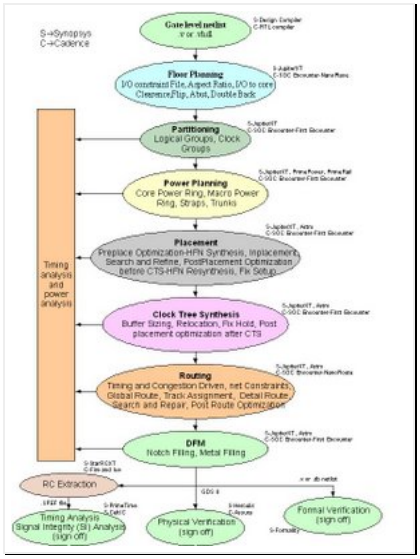
- Increased net length can accumulate more charges while manufacturing of the device due to ionisation process. If this net is connected to gate of the MOSFET it can damage dielectric property of the gate and gate may conduct causing damage to the MOSFET. This is antenna problem.
- Decrease the length of the net by providing more vias and layer jumping.
- Insert antenna diode.

How delays vary with different PVT conditions? Show the graph.

- P increase->dealy increase
- P decrease->delay decrease

- V increase->delay decrease
- V decrease->delay increase
- T increase->delay increase
- T decrease->delay decrease

Explain the flow of physical design and inputs and outputs for each step in flow.
Physical Design Flow



The physical design flow is generally explained in the Figure (1.). In each section of the flow EDA tools available from the two main EDA companies-Synopsys and Cadence is also listed. In each and every step of the flow timing and power analysis can be carried out. If timing and power requirements are not met then either the whole flow has to be re-exercised or going back one or two steps and optimizing the design or incremental optimization may meet the requirements

What is cell delay and net delay?

- **Gate delay**
- Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]
- Gate delay =function of(i/p transition time, Cnet+Cpin).
- Cell delay is also same as Gate delay.
- **Cell delay**
- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.
- Intrinsic delay
- Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.
- It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.
- This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.
- **Net Delay (or wire delay)**
- The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.
- It is due to the finite resistance and capacitance of the net.It is also known as wire delay.
- Wire delay =fn(Rnet , Cnet+Cpin)

What are delay models and what is the difference between them?

- Linear Delay Model (LDM)
- Non Linear Delay Model (NLDM)

What is wire load model?

- Wire load model is NLDM which has estimated R and C of the net.

Why higher metal layers are preferred for Vdd and Vss?

- Because it has less resistance and hence leads to less IR drop.

What is logic optimization and give some methods of logic optimization.

- Upsizing
- Downsizing
- Buffer insertion
- Buffer relocation
- Dummy buffer placement

What is the significance of negative slack?

- negative slack==> there is setup voilation==> deisgn can fail

What is signal integrity? How it affects Timing?

- IR drop, Electro Migration (EM), Crosstalk, Ground bounce are signal integrity issues.
- If Idrop is more==>delay increases.
- crosstalk==>there can be setup as well as hold voilation.

What is IR drop? How to avoid? How it affects timing?

- There is a resistance associated with each metal layer. This resistance consumes power causing voltage drop i.e.IR drop.
- If IR drop is more==>delay increases.

What is EM and it effects?

- Due to high current flow in the metal atoms of the metal can displaced from its origial place. When it happens in larger amount the metal can open or bulging of metal layer can happen. This effect is known as Electro Migration.
- Affects: Either short or open of the signal line or power line.

What are types of routing?

- Global Routing
- Track Assignment
- Detail Routing

What is latency? Give the types?

- **Source Latency**
- It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".
- Delay from clock source to beginning of clock tree (i.e. clock definition point).
- The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.
- **Network latency**
- It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".
- The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

What is track assignment?

- Second stage of the routing wherein particular metal tracks (or layers) are assigned to the signal nets.

What is congestion?

- If the number of routing tracks available for routing is less than the required tracks then it is known as congestion.

Whether congestion is related to placement or routing?

- Routing

What are clock trees?

- Distribution of clock from the clock source to the sync pin of the registers.

What are clock tree types?

- H tree, Balanced tree, X tree, Clustering tree, Fish bone

What is cloning and buffering?

- Cloning is a method of optimization that decreases the load of a heavily loaded cell by replicating the cell.
- Buffering is a method of optimization that is used to insert beffers in high fanout nets to decrease the dealy.

ASIC

Different Types of Delays in ASIC or VLSI design

- **Source Delay/Latency**
- **Network Delay/Latency**
- **Insertion Delay**
- **Transition Delay/Slew: Rise time, fall time**
- **Path Delay**
- **Net delay, wire delay, interconnect delay**
- **Propagation Delay**
- **Phase Delay**
- **Cell Delay**
- **Intrinsic Delay**
- **Extrinsic Delay**
- **Input Delay**
- **Output Delay**
- **Exit Delay**
- **Latency (Pre/post CTS)**
- **Uncertainty (Pre/Post CTS)**

- **Unateness: Positive unateness, negative unateness**
- **Jitter: PLL jitter, clock jitter**

Gate delay

- Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]
- Gate delay =function of(i/p transition time, Cnet+Cpin).
- Cell delay is also same as Gate delay.

Source Delay (or Source Latency)

- It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".
- Delay from clock source to beginning of clock tree (i.e. clock definition point).
- The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.

Network Delay(latency)

- It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".
- The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

Insertion delay

- The delay from the clock definition point to the clock pin of the register.

Transition delay

- It is also known as "Slew". It is defined as the time taken to change the state of the signal. Time taken for the transition from logic 0 to logic 1 and vice versa . or Time taken by the input signal to rise from 10%(20%) to the 90% (80%) and vice versa.
- Transition is the time it takes for the pin to change state.

Slew

- Rate of change of logic.See Transition delay.
- Slew rate is the speed of transition measured in volt / ns.

Rise Time

- Rise time is the difference between the time when the signal crosses a low threshold to the time when the signal crosses the high threshold. It can be absolute or percent.
- Low and high thresholds are fixed voltage levels around the mid voltage level or it can be either 10% and 90% respectively or 20% and 80% respectively. The percent levels are converted to absolute voltage levels at the time of measurement by calculating percentages from the difference between the starting voltage level and the final settled voltage level.

Fall Time

- Fall time is the difference between the time when the signal crosses a high threshold to the time when the signal crosses the low threshold.
- The low and high thresholds are fixed voltage levels around the mid voltage level or it can be either 10% and 90% respectively or 20% and 80% respectively. The percent levels are converted to absolute voltage levels at the time of measurement by calculating percentages from the difference between the starting voltage level and the final settled voltage level.
- For an ideal square wave with 50% duty cycle, the rise time will be 0.For a symmetric triangular wave, this is reduced to just 50%.
- The rise/fall definition is set on the meter to 10% and 90% based on the linear power in Watts. These points translate into the -10 dB and -0.5 dB points in log mode (10 log 0.1) and (10 log 0.9). The rise/fall time values of 10% and 90% are calculated based on an algorithm, which looks at the mean power above and below the 50% points of the rise/fall times
- **Path delay**
- Path delay is also known as pin to pin delay. It is the delay from the input pin of the cell to the output pin of the cell.

Net Delay (or wire delay)

- The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.
- It is due to the finite resistance and capacitance of the net.It is also known as wire delay.
- Wire delay =fn(Rnet , Cnet+Cpin)

Propagation delay

- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.
- This is the time required for a signal to propagate through a gate or net. For gates it is the time it takes for a event at the gate input to affect the gate output.
- For net it is the delay between the time a signal is first applied to the net and the time it reaches other devices connected to that net.
- It is taken as the average of rise time and fall time i.e. Tpd= (Tphl+Tplh)/2.

Phase delay

- Same as insertion delay

Cell delay

- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.

Intrinsic delay

- Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.
- It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.
- This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.

Extrinsic delay

- Same as wire delay, net delay, interconnect delay, flight time.
- Extrinsic delay is the delay effect that associated to with interconnect. output pin of the cell to the input pin of the next cell.

Input delay

- Input delay is the time at which the data arrives at the input pin of the block from external circuit with respect to reference clock.

Output delay

- Output delay is time required by the external circuit before which the data has to arrive at the output pin of the block with respect to reference clock.

Exit delay

- It is defined as the delay in the longest path (critical path) between clock pad input and an output. It determines the maximum operating frequency of the design.

Latency (pre/post cts)

- Latency is the summation of the Source latency and the Network latency. Pre CTS estimated latency will be considered during the synthesis and after CTS propagated latency is considered.

Uncertainty (pre/post cts)

- Uncertainty is the amount of skew and the variation in the arrival clock edge. Pre CTS uncertainty is clock skew and clock Jitter. After CTS we can have some margin of skew + Jitter.

Unateness

- A function is said to be unate if the rise transition on the positive unate input variable causes the ouput to rise or no change and vice versa.
- Negative unateness means cell output logic is inverted version of input logic. eg. In inverter having input A and output Y, Y is -ve unate w.r.to A. Positive unate means cell output logic is same as that of input.
- These +ve ad -ve unateness are constraints defined in library file and are defined for output pin w.r.to some input pin.
- A clock signal is positive unate if a rising edge at the clock source can only cause a rising edge at the register clock pin, and a falling edge at the clock source can only cause a falling edge at the register clock pin.
- A clock signal is negative unate if a rising edge at the clock source can only cause a falling edge at the register clock pin, and a falling edge at the clock source can only cause a rising edge at the register clock pin. In other words, the clock signal is inverted.
- A clock signal is not unate if the clock sense is ambiguous as a result of non-unate timing arcs in the clock path. For example, a clock that passes through an XOR gate is not unate because there are nonunate arcs in the gate. The clock sense could be either positive or negative, depending on the state of the other input to the XOR gate.

Jitter

- The short-term variations of a signal with respect to its ideal position in time.
- Jitter is the variation of the clock period from edge to edge. It can varry +/- jitter value.
- From cycle to cycle the period and duty cycle can change slightly due to the clock generation circuitry. This can be modeled by adding uncertainty regions around the rising and falling edges of the clock waveform.

Sources of Jitter Common sources of jitter include:

- Internal circuitry of the phase-locked loop (PLL)
- Random thermal noise from a crystal
- Other resonating devices
- Random mechanical noise from crystal vibration
- Signal transmitters
- Traces and cables
- Connectors
- Receivers

Skew

- The difference in the arrival of clock signal at the clock pin of different flops.
- Two types of skews are defined: Local skew and Global skew.

Local skew

- The difference in the arrival of clock signal at the clock pin of related flops.

Global skew

- The difference in the arrival of clock signal at the clock pin of non related flops.
- Skew can be positive or negative.
- When data and clock are routed in same direction then it is **Positive skew.**
- When data and clock are routed in opposite then it is **negative skew.**

Recovery Time

- Recovery specifies the minimum time that an asynchronous control input pin must be held stable after being de-asserted and before the next clock (active-edge) transition.
- Recovery time specifies the time the inactive edge of the asynchronous signal has to arrive before the closing edge of the clock.
- Recovery time is the minimum length of time an asynchronous control signal (eg.preset) must be stable before the next active clock edge. The recovery slack time calculation is similar to the clock setup slack time calculation, but it applies asynchronous control signals.

Equation 1:

- Recovery Slack Time = Data Required Time â€" Data Arrival Time
- Data Arrival Time = Launch Edge + Clock Network Delay to Source Register + Tclkq+ Register to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register =Tsetup

If the asynchronous control is not registered, equations shown in Equation 2 is used to calculate the recovery slack time.
Equation 2:

- Recovery Slack Time = Data Required Time â€" Data Arrival Time
- Data Arrival Time = Launch Edge + Maximum Input Delay + Port to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register Delay+Tsetup
- If the asynchronous reset signal is from a port (device I/O), you must make an Input Maximum Delay assignment to the asynchronous reset pin to perform recovery analysis on that path.

Removal Time

- Removal specifies the minimum time that an asynchronous control input pin must be held stable before being de-asserted and after the previous clock (active-edge) transition.
- Removal time specifies the length of time the active phase of the asynchronous signal has to be held after the closing edge of clock.
- Removal time is the minimum length of time an asynchronous control signal must be stable after the active clock edge. Calculation is similar to the clock hold slack calculation, but it applies asynchronous control signals. If the asynchronous control is registered, equations shown in Equation 3 is used to calculate the removal slack time.
- If the recovery or removal minimum time requirement is violated, the output of the sequential cell becomes uncertain. The uncertainty can be caused by the value set by the resetbar signal or the value clocked into the sequential cell from the data input.

Equation 3

- Removal Slack Time = Data Arrival Time â€" Data Required Time
- Data Arrival Time = Launch Edge + Clock Network Delay to Source Register + Tclkq of Source Register + Register to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register + Thold
- If the asynchronous control is not registered, equations shown in Equation 4 is used to calculate the removal slack time.

Equation 4

- Removal Slack Time = Data Arrival Time â€" Data Required Time
- Data Arrival Time = Launch Edge + Input Minimum Delay of Pin + Minimum Pin to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register +Thold
- If the asynchronous reset signal is from a device pin, you must specify the Input Minimum Delay constraint to the asynchronous reset pin to perform a removal analysis on this path.

What is the difference between soft macro and hard macro?

- **What is the difference between hard macro, firm macro and soft macro?**

or

- **What are IPs?**
- Hard macro, firm macro and soft macro are all known as IP (Intellectual property). They are optimized for power, area and performance. They can be purchased and used in your ASIC or FPGA design implementation flow. Soft macro is flexible for all type of ASIC implementation. Hard macro can be used in pure ASIC design flow, not in FPGA flow. Before bying any IP it is very important to evaluate its advantages and disadvantages over each other, hardware compatibility such as I/O standards with your design blocks, reusability for other designs.

Soft macros

- Soft macros are in synthesizable RTL.
- Soft macros are more flexible than firm or hard macros.
- Soft macros are not specific to any manufacturing process.
- Soft macros have the disadvantage of being somewhat unpredictable in terms of performance, timing, area, or power.
- Soft macros carry greater IP protection risks because RTL source code is more portable and therefore, less easily protected than either a netlist or physical layout data.
- From the physical design perspective, soft macro is any cell that has been placed and routed in a placement and routing tool such as Astro. (This is the definition given in Astro Rail user manual !)
- Soft macros are editable and can contain standard cells, hard macros, or other soft macros.

Firm macros

- Firm macros are in netlist format.
- Firm macros are optimized for performance/area/power using a specific fabrication technology.
- Firm macros are more flexible and portable than hard macros.
- Firm macros are predictive of performance and area than soft macros.

Hard macro

- Hard macros are generally in the form of hardware IPs (or we termed it as hardwre IPs !).
- Hard macos are targeted for specific IC manufacturing technology.
- Hard macros are block level designs which are silicon tested and proved.
- Hard macros have been optimized for power or area or timing.
- In physical design you can only access pins of hard macros unlike soft macros which allows us to manipulate in different way.
- You have freedom to move, rotate, flip but you can't touch anything inside hard macros.
- Very common example of hard macro is memory. It can be any design which carries dedicated single functionality (in general).. for example it can be a MP4 decoder.
- Be aware of features and characteristics of hard macro before you use it in your design... other than power, timing and area you also should know pin properties like sync pin, I/O standards etc
- LEF, GDS2 file format allows easy usage of macros in different tools.

From the physical design (backend) perspective:

- Hard macro is a block that is generated in a methodology other than place and route (i.e. using full custom design methodology) and is brought into the physical design database (eg. Milkyway in Synopsys; Volcano in Magma) as a GDS2 file.

What is the difference between FPGA and CPLD?

FPGA-Field Programmable Gate Array and CPLD-Complex Programmable Logic Device-- both are programmable logic devices made by the same companies with different characteristics.

"A Complex Programmable Logic Device (CPLD) is a Programmable Logic Device with complexity between that of PALs (Programmable Array Logic) and FPGAs, and architectural features of both. The building block of a CPLD is the macro cell, which contains logic implementing disjunctive normal form expressions and more specialized logic operations".

Architecture

Granularity is the biggest difference between CPLD and FPGA.

FPGA are "fine-grain" devices. That means that they contain hundreds of (up to 100000) of tiny blocks (called as LUT or CLBs etc) of logic with flip-flops, combinational logic and memories.FPGAs offer much higher complexity, up to 150,000 flip-flops and large number of gates available.

CPLDs typically have the equivalent of thousands of logic gates, allowing implementation of moderately complicated data processing devices. PALs typically have a few hundred gate equivalents at most, while FPGAs typically range from tens of thousands to several million.

CPLD are "coarse-grain" devices. They contain relatively few (a few 100's max) large blocks of logic with flip-flops and combinational logic. CPLDs based on AND-OR structure.

CPLD's have a register with associated logic (AND/OR matrix). CPLD's are mostly implemented in control applications and FPGA's in datapath applications. Because of this course grained architecture, the timing is very fixed in CPLDs.

- FPGA are RAM based. They need to be "downloaded" (configured) at each power-up. CPLD are EEPROM based. They are active at power-up i.e. as long as they've been programmed at least once.

FPGA needs boot ROM but CPLD does not. In some systems you might not have enough time to boot up FPGA then you need CPLD+FPGA.

Generally, the CPLD devices are not volatile, because they contain flash or erasable ROM memory in all the cases. The FPGA are volatile in many cases and hence they need a configuration memory for working. There are some FPGAs now which are nonvolatile. This distinction is rapidly becoming less relevant, as several of the latest FPGA products also offer models with embedded configuration memory.

- The characteristic of non-volatility makes the CPLD the device of choice in modern digital designs to perform 'boot loader' functions before handing over control to other devices not having this capability. A good example is where a CPLD is used to load configuration data for an FPGA from non-volatile memory.
- Because of coarse-grain architecture, one block of logic can hold a big equation and hence CPLD have a faster input-to-output timings than FPGA.

Features

- FPGA have special routing resources to implement binary counters,arithmetic functions like adders, comparators and RAM. CPLD don't have special features like this.
- FPGA can contain very large digital designs, while CPLD can contain small designs only.The limited complexity (<500>
- ***Speed***: CPLDs offer a single-chip solution with fast pin-to-pin delays, even for wide input functions. Use CPLDs for small designs, where "instant-on", fast and wide decoding, ultra-low idle power consumption, and design security are important (e.g., in battery-operated equipment).
- ***Security***: In CPLD once programmed, the design can be locked and thus made secure. Since the configuration bitstream must be reloaded every time power is re-applied, design security in FPGA is an issue.
- ***Power***: The high static (idle) power consumption prohibits use of CPLD in battery-operated equipment. FPGA idle power consumption is reasonably low, although it is sharply increasing in the newest families.
- ***Design flexibility***: FPGAs offer more logic flexibility and more sophisticated system features than CPLDs: clock management, on-chip RAM, DSP functions, (multipliers), and even on-chip microprocessors and Multi-Gigabit Transceivers.These benefits and opportunities of dynamic reconfiguration, even in the end-user system, are an important advantage.
- Use FPGAs for larger and more complex designs.
- FPGA is suited for timing circuit becauce they have more registers , but CPLD is suited for control circuit because they have more combinational circuit. At the same time, If you synthesis the same code for FPGA for many times, you will find out that each timing report is different. But it is different in CPLD synthesis, you can get the same result.

As CPLDs and FPGAs become more advanced the differences between the two device types will continue to blur. While this trend may appear to make the two types more difficult to keep apart, the architectural advantage of CPLDs combining low cost, non-volatile configuration, and macro cells with predictable timing characteristics will likely be sufficient to maintain a product differentiation for the foreseeable future.

What is the difference between FPGA and ASIC?

This question is very popular in VLSI fresher interviews. It looks simple but a deeper insight into the subject reveals the fact that there are lot of thinks to be understood !! So here is the answer.

FPGA vs. ASIC

- Difference between ASICs and FPGAs mainly depends on costs, tool availability, performance and design flexibility. They have their own pros and cons but it is designers responsibility to find the advantages of the each and use either FPGA or ASIC for the product. However, recent developments in the FPGA domain are narrowing down the benefits of the ASICs.

FPGA

- **Field Programable Gate Arrays**

FPGA Design Advantages

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ASIC

- **Application Specific Intergrated Circiut**

ASIC Design Advantages

- ***Cost....cost....cost....Lower unit costs:*** For very high volume designs costs comes out to be very less. Larger volumes of ASIC design proves to be cheaper than implementing design using FPGA.
- ***Speed...speed...speed....ASICs are faster than FPGA:*** ASIC gives design flexibility. This gives enoromous opportunity for speed optimizations.
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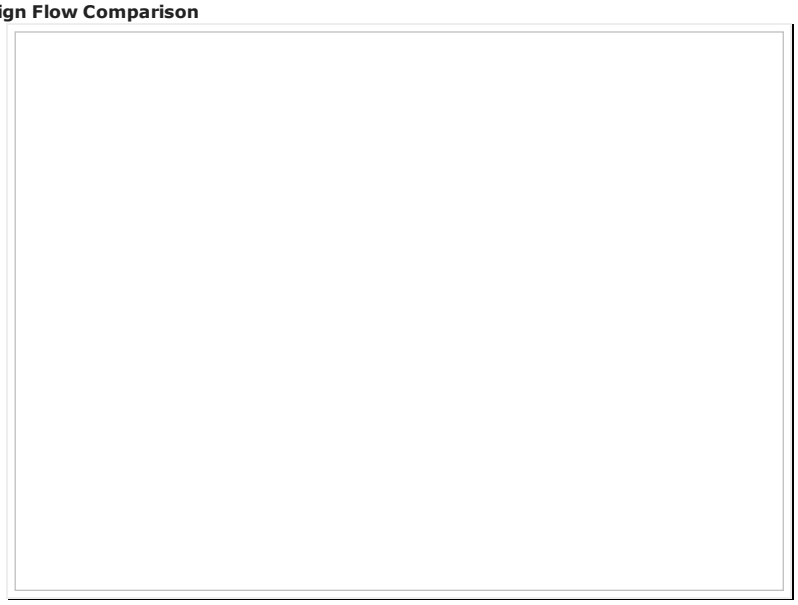
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FPGA vs. ASIC Design Flow Comparison



ASIC Design Check List

Silicon Process and Library Characteristics

- What exact process are you using?
- How many layers can be used for this design?
- Are the Cross talk Noise constraints, Xtalk Analysis configuration, Cell EM & Wire EM available?

Design Characteristics

- What is the design application?
- Number of cells (placeable objects)?
- Is the design Verilog or VHDL?
- Is the netlist flat or hierarchical?
- Is there RTL available?
- Is there any datapath logic using special datapath tools?
- Is the DFT to be considered?
- Can scan chains be reordered?
- Is memory BIST, boundary scan used on this design?
- Are static timing analysis constraints available in SDC format?

Clock Characteristics

- How many clock domains are in the design?
- What are the clock frequencies?
- Is there a target clock skew, latency or other clock requirements?
- Does the design have a PLL?
- If so, is it used to remove clock latency?
- Is there any I/O cell in the feedback path?
- Is the PLL used for frequency multipliers?
- Are there derived clocks or complex clock generation circuitry?
- Are there any gated clocks?
- If yes, do they use simple gating elements?
- Is the gate clock used for timing or power?
- For gated clocks, can the gating elements be sized for timing?
- Are you muxing in a test clock or using a JTAG clock?
- Available cells for clock tree?
- Are there any special clock repeaters in the library?
- Are there any EM, slew or capacitance limits on these repeaters?
- How many drive strengths are available in the standard buffers and inverters?
- Do any of the buffers have balanced rise and fall delays?
- Any there special requirements for clock distribution?
- Will the clock tree be shielded? If so, what are the shielding requirements?

Floorplan and Package Characteristics

- Target die area?
- Does the area estimate include power/signal routing?
- What gates/mm2 has been assumed?

- Number of routing layers?
- Any special power routing requirements?
- Number of digital I/O pins/pads?
- Number of analog signal pins/pads?
- Number of power/ground pins/pads?
- Total number of pins/pads and Location?
- Will this chip use a wire bond package?
- Will this chip use a flip-chip package?
- If Yes, is it I/O bump pitch? Rows of bumps? Bump allocation?Bump pad layout guide?
- Have you already done floorplanning for this design?
- If yes, is conformance to the existing floorplan required?
- What is the target die size?
- What is the expected utilization?
- Please draw the overall floorplan ?
- Is there an existing floorplan available in DEF?
- What are the number and type of macros (memory, PLL, etc.)?
- Are there any analog blocks in the design?
- What kind of packaging is used? Flipchip?
- Are the I/Os periphery I/O or area I/O?
- How many I/Os?
- Is the design pad limited?
- Power planning and Power analysis for this design?
- Are layout databases available for hard macros ?
- Timing analysis and correlatio?
- Physical verification ?

Data Input

- Library information for new library
- .lib for timing information
- GDSII or LEF for library cells including any RAMs
- RTL in Verilog/VHDL format
- Number of logical blocks in the RTL
- Constraints for the block in SDC
- Floorplan information in DEF
- I/O pin location
- Macro locations

ASIC General

General ASIC questions are posted here. More questions related to different catagories of ASICs can be found at respective sections.

- What are the differences between PALs, PLAs, FPGAs, ASICs and PLDs?
- In system with insufficient hold time, will slowing down the clock help?
- In system with insufficient setup time, will slowing down the clock help?
- Why would a testbench not have pins (port) on it?
- When declaring a flip flop, why would not you declare its output value in the port statement?
- Give 2 advantages of using a script to build a chip?
- A "tri state " bus is directly connected to a set of CMOS input buffers. No other wires or components are attached to the bus wires. Upon observation we can find that under certain conditions, this circuit is consuming considerable power. Why it is so? Is circuit correct? If not, how to correct?
- Is Verilog (or that matter any HDL) is a concurrent or sequential language?
- What is the function of sensitivity list?
- A mealy –type state machine is coded using D-type rising edge flip flops. The reset and clock signals are in the sensitivity list but with one of the next state logic input signals have been left out of the sensitivity list. Explain what happens when the state machine is simulated? Will the state machine be synthesized correctly?
- A moore –type state machine is coded using D-type rising edge flip flops. The reset and clock signals are in the sensitivity list but with one of the next state logic input signals have been left out of the sensitivity list. Explain what happens when the state machine is simulated? Will the state machine be synthesized correctly?
- What type of delay is most like a infinite bandwidth transmission line?
- Define metastability.
- When does metastability occur?
- Give one example of a situation where metastability could occur.
- Give two ways metastability could manifest itself in a state machine.
- What is MTBF?
- Does MTBF give the time until the next failure occurs?
- Give 3 ways in which to reduce the chance of metastable failure.
- Give 2 advantages of using a synchronous reset methodology.
- Give 2 disadvantages of using a synchronous reset methodology.
- Give 2 advantages of using an asynchronous reset methodology.
- Give 2 disadvantages of using an asynchronous reset methodology.
- What are the two most fundamental inputs (files) to the synthesis tool?
- What are two important steps in synthesis? What happens in those steps?

- What are the two major output (files) from the synthesis process?
- Name the fundamental 3 operating consitions that determine (globally) the delay characteristics of CMOS gates. For each how they affect gate delay?
- For a single gate, with global gating conditions held constant , what 3 delay coefficients effect total gate delay? Which is the most sensitive to circuit topology?

FPGA.

What is the difference between FPGA and CPLD?

FPGA-Field Programmable Gate Array and CPLD-Complex Programmable Logic Device-- both are programmable logic devices made by the same companies with different characteristics.

- "A Complex Programmable Logic Device (CPLD) is a Programmable Logic Device with complexity between that of PALs (Programmable Array Logic) and FPGAs, and architectural features of both. The building block of a CPLD is the macro cell, which contains logic implementing disjunctive normal form expressions and more specialized logic operations".
- This is what Wiki defines.....!!

Architecture

- Granularity is the biggest difference between CPLD and FPGA.
- FPGA are "fine-grain" devices. That means that they contain hundreds of (up to 100000) of tiny blocks (called as LUT or CLBs etc) of logic with flip-flops, combinational logic and memories.FPGAs offer much higher complexity, up to 150,000 flip-flops and large number of gates available.
- CPLDs typically have the equivalent of thousands of logic gates, allowing implementation of moderately complicated data processing devices. PALs typically have a few hundred gate equivalents at most, while FPGAs typically range from tens of thousands to several million.

CPLD are "coarse-grain" devices. They contain relatively few (a few 100's max) large blocks of logic with flip-flops and combinational logic. CPLDs based on AND-OR structure.

CPLD's have a register with associated logic (AND/OR matrix). CPLD's are mostly implemented in control applications and FPGA's in datapath applications. Because of this course grained architecture, the timing is very fixed in CPLDs.

FPGA are RAM based. They need to be "downloaded" (configured) at each power-up. CPLD are EEPROM based. They are active at power-up i.e. as long as they've been programmed at least once.

FPGA needs boot ROM but CPLD does not. In some systems you might not have enough time to boot up FPGA then you need CPLD+FPGA.

- Generally, the CPLD devices are not volatile, because they contain flash or erasable ROM memory in all the cases. The FPGA are volatile in many cases and hence they need a configuration memory for working. There are some FPGAs now which are nonvolatile. This distinction is rapidly becoming less relevant, as several of the latest FPGA products also offer models with embedded configuration memory.
- The characteristic of non-volatility makes the CPLD the device of choice in modern digital designs to perform 'boot loader' functions before handing over control to other devices not having this capability. A good example is where a CPLD is used to load configuration data for an FPGA from non-volatile memory.
- Because of coarse-grain architecture, one block of logic can hold a big equation and hence CPLD have a faster input-to-output timings than FPGA.

Features

- FPGA have special routing resources to implement binary counters,arithmetic functions like adders, comparators and RAM. CPLD don't have special features like this.
- FPGA can contain very large digital designs, while CPLD can contain small designs only.The limited complexity (<500>
- ***Speed***: CPLDs offer a single-chip solution with fast pin-to-pin delays, even for wide input functions. Use CPLDs for small designs, where "instant-on", fast and wide decoding, ultra-low idle power consumption, and design security are important (e.g., in battery-operated equipment).
- ***Security***: In CPLD once programmed, the design can be locked and thus made secure. Since the configuration bitstream must be reloaded every time power is re-applied, design security in FPGA is an issue.
- ***Power***: The high static (idle) power consumption prohibits use of CPLD in battery-operated equipment. FPGA idle power consumption is reasonably low, although it is sharply increasing in the newest families.

Design flexibility: FPGAs offer more logic flexibility and more sophisticated system features than CPLDs: clock management, on-chip RAM, DSP functions, (multipliers), and even on-chip microprocessors and Multi-Gigabit Transceivers.These benefits and opportunities of dynamic reconfiguration, even in the end-user system, are an important advantage.

- Use FPGAs for larger and more complex designs.
- FPGA is suited for timing circuit becauce they have more registers , but CPLD is suited for control circuit because they have more combinational circuit. At the same time, If you synthesis the same code for FPGA for many times, you will find out that each timing report is different. But it is different in CPLD synthesis, you can get the same result.

As CPLDs and FPGAs become more advanced the differences between the two device types will continue to blur. While this trend may appear to make the two types more difficult to keep apart, the architectural advantage of CPLDs combining low cost, non-volatile configuration, and macro cells with predictable timing characteristics will likely be sufficient to maintain a product differentiation for the foreseeable future.

What is the difference between FPGA and ASIC?

- This question is very popular in VLSI fresher interviews. It looks simple but a deeper insight into the subject reveals the fact that there are lot of thinks to be understood !! So here is the answer.

FPGA vs. ASIC

- Difference between ASICs and FPGAs mainly depends on costs, tool availability, performance and design flexibility. They have their own pros and cons but it is designers responsibility to find the advantages of the each and use either FPGA or ASIC for the product. However, recent developments in the FPGA domain are narrowing down the benefits of the ASICs.

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FPGA Interview Questions

· What is minimum and maximum frequency of DCM in spartan-3 series FPGA?
· List some of constraints you used and their purpose during your design?
· What is the size of bitmap with changing gate count?

- What are different types of FPGA programming modes? How to change from one to another?
- List out some important features of FPGA.
- List out some of synthesizable and non synthesizable constructs?
- Draw general structure of FPGA?
- What is the difference between FPGA and CPLD?
- What is DCM? Why they are used?
- Draw FPGA design flow. Explain each step. What is input and output from each step?
- What is slice, CLB, LUT?
- Is it possible to configure CLB as RAM?
- What is purpose of a constraint file? What is its extension?
- How you will choose an FPGA?
- How clock is routed through out FPGA?
- What are difference between PLL and DLL ?
- What is soft processor?
- What is hard processor?



2 comments:



navin kumar February 11, 2014 at 12:04 PM

This comment has been removed by the author.

Reply



anagha ghosh April 30, 2014 at 10:52 AM

Good Work Sir !!

Reply

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